DYNAM3D: DYNAMIC LAYERED 3D TOKENS EMPOWER VLM FOR NAVIGATION

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1. Summary of the Paper

Wang et al. (2025) present the concept of Dynam3D, the dynamic layering 3D representation framework that is expected to enhance Vision-and-Language Navigation (VLN) in real-life settings. The conventional VLN systems typically use video-based vision-language models that operate on image frames in sequence, so it is difficult to get the right spatial geometry of 3D objects, a long-term environment memory and dynamic adaptation. To overcome these drawbacks, Dynam3D provides a hierarchical 3D scene coding in the form of tokens (patch-, instance-, zone-level) that represent the fine-scale visual detail of the target image as well as the large-scale spatial context. The framework initially derives 2D patch characteristics in RGB images by utilizing a CLIP encoder, and then projects them into 3D space by utilizing depth maps and camera poses. A frustum-culling mechanism is used to dynamically update these features and eliminate old information.

FastSAM instance segmentation produces object-level representations which are combined across many views through a trained merging discriminator so that 3D instances are consistent. The model also collimates instances into space regions in order to represent upper environmental arrangements. A 3D Vision-Language Model with LLaVA-Phi-3-mini is used to predict navigation actions with these hierarchical 3D tokens together with the action history and navigation instructions. The VLN benchmark experiments, which involve R2R-CE, REVERIE-CE and NavRAG-CE, prove that Dynam3D performs with the state-of-the-art speeds, and at the same time, with high efficiency in real-time navigation.

2. Critical Appraisal

2.1 Theoretical Foundations

The Dynam3D framework is based on some major principles of deep learning that allow it to make use of multimodal understanding and spatial reasoning. One of them is known as multimodal representation learning, which aligns visual and textual data in a common embedding space. The model relies on CLIP embeddings, in which visual features that are generated out of image patches and text descriptions are projected into a shared semantic space. The model uses contrastive learning to increase the similarity between similar pairs of images and texts and minimize the distance between unrelated pairs (Li et al. 2026). Such alignment enables the navigation system to match language instructions, including names of objects or description of rooms with visual objects in the environment.

Hierarchical scene representation is another concept that is important. The Dynam3D divides the environment into three levels, patch tokens, instance tokens and zone tokens. Patch tokens

are fine-placed visual details of CLIP, the instance tokens are the patches of an object, and the zone tokens are the bigger spatial units like rooms (Zhang et al. 2025). The hierarchy is similar to feature abstraction in convolutional neural networks and transformer models, where low-level features are iteratively aggregated to create higher-level semantic representations.

By using the LLaVA-Phi-3-mini model, the system uses transformer-based reasoning, which handles multimodal tokens with self-attention mechanisms learning the relationship between visual tokens and instructions as well as previous actions. Contrastive losses and feature distillation are used to obtain the optimum, and backpropagation is used to optimize the representations at each hierarchical level.

2.2 Methodological Evaluation

Dynam3D experimental design is considered in three popular Vision-and-Language Navigation benchmarks: R2R-CE, REVERIE-CE, and NavRAG-CE. These datasets offer natural indoor navigation environments where agents need to use natural language instructions to get to the target locations. R2R-CE is step by step navigation guide, whereas REVERIE-CE has an abstract description of destinations and NavRAG-CE has the requirement of reasoning over complex user demands. Even though these datasets do offer a convenient testbed to multimodal navigation systems, they are mostly made up of indoor settings based on datasets like Matterport3D, and thus the environment might be limited in terms of diversity (Habibpour and Afghah, 2025). Moreover, although the paper presents experiments in dynamic environments, the benchmark datasets are mostly an indoor environment, which can add to the lack of realism of assessing dynamic environment models.

The authors make comparisons of Dynam3D with some of the robust baselines such as NaVid, g3D-LF and Uni-NaVid that are the state of art approaches to make use of video based or feature field representation. These comparisons are informative, but they are questionable in terms of fairness since the model settings and training conditions may differ. As an example, Uni-NaVid has a bigger language model (7B parameters) compared to Dynam3D (3.8B parameters). Also, the differences in training data or preprocessing choices between methods can affect performance, and thus the direct comparisons are not so controlled.

Navigation Error (NE), Success rate (SR) and Success weighted by Path Length (SPL) are used as a measure of model performance. The NE is the distance between the agent and the predicted and target position, SR is an evaluation of whether the agent manages to reach the goal, and SPL is an evaluation of the efficiency of navigation by punishing unnecessarily long distances (Yin et al. 2024). These measurements, although they can reflect the precision and efficiency

of navigation, are not yet a direct measure of reasoning ability or semantic interpretation of commands.

The ablation experiments examine the roles of instance representations, zone representations and subspace contrastive learning. The significance of these experiments is that when the components are removed it will result in performance degradation. Nevertheless, further ablations of the memory scalability, accuracy of memory segmentation or computational trade-offs would contribute to the evaluation.

2.3 Drawbacks of the Model

The Dynam3D framework has a number of limitations that could potentially influence its implementation, despite its high performance. Computational cost is one of the major problems. This system combines various heavy components, such as CLIP to extract visual features, FastSAM to segment instances, and LLaVi-Phi-3-mini vision-language component to reason. Even though the authors do mention near real-time inference, these modules are very resource-intensive when it comes to the usage of the GPUs. That might restrict the applications on mobile robots with limited onboard hardware and might consume more energy when a long-range direction is required (Wang et al. 2025).

The other constraint is the scalability of memory. Dynam3D also stores extensive quantities of 3D patch features to create the feature field and provide memory on the scene. The size of the environments searched by the agent may increase substantially, which may cause the stored patches to grow and consequently increase the amount of memory and time taken to perform update operations. The representation can be scaled to very big or long-duration environments with no efficient compression or pruning strategies.

The model is also based on RGB-D, where depth sensors are necessary to project 2D features properly in 3D space. Nevertheless, several working robotic platforms are based solely on monocular cameras, which can limit the applicability of the method. Moreover, the mistakes of the FastSAM stage of instance segmentation can be carried into the pipeline, leading to false object merging as well as faulty representation of instances.

3. Proposal for Improvement

3.1 Need for Improvement

Although there is excellent performance of Dynam3D in terms of vision-and-language navigation, a number of limitations can be pointed out to spur future improvement. Computational complexity of the framework is one of the issues. It makes use of several bulky parts such as CLIP to extract features, FastSAM to segment instances, and the reasoning of a

large vision-language model. Even though the authors describe almost real-time performance, this pipeline may be computationally intense even on robotic platforms with a limited onboard hardware. Limiting the computational costs and preserving the semantic knowledge is thus a valuable improvement direction (Lin et al. 2023).

The other constraint is associated with world spatial reasoning. Although hierarchical patch-instance-zone representation gives us systematic environmental knowledge, the instance/zone interaction is mainly influenced by spatial integration as opposed to explicit relational modelling. This can make the system have difficulties in representing intricate physical relations amongst objects or areas which are frequently needed to interpret navigation directions like the chair beside the table in the dwelling room. The enhancement of the relational reasoning that exists between objects and regions might thus improve the performance of navigation (Yuan et al. 2025).

Also, the framework relies heavily on instance segmentation accuracy of FastSAM. Segmentation errors can be spread up the hierarchy of the representation resulting in the wrong merging of instances and unreliable semantic representation.

3.2 Extension Method Proposed

In order to overcome the shortcomings of the Dynam3D framework, this proposal proposes a more realistic graph-based scene representation method based on Graph Neural Networks (GNNs) to improve spatial reasoning and object relationship modelling. The original architecture uses hierarchical patches, instances and zone tokens to capture the environment with zone-level features being a result of aggregating the surrounding instances within fixed spatial cubes. Although such a method captures rough spatial information, relationships between objects or regions are not explicitly modelled in it, and these are often very important in the interpretation of navigation instructions that involve spatial relationships (Lin et al. 2023).

The suggested technique substitutes the zone aggregation module with a graphical representation of space. In this architecture, every occurrence of detected objects is modelled as a node in a graph, which has the semantic features of the aggregated patch representations. The spatial relationships between nodes are represented using edges between nodes, including distance, relative orientation and co-location in the environment. This structure is processed by a Graph Neural Network and contextual information between objects is propagated, enabling the model to make reasoning about relationships, like adjacency or containment (Yin et al. 2024).

The modified pipeline would then change instance tokens into graph nodes and subsequently run through a series of GNN layers after which those are fed into the 3D Vision-Language Model. The relational mechanism of reasoning allows to interpret the instructions that involve relationships between objects more accurately and retain the hierarchical representation of the Dynam3D framework present.

3.3 Architecture Modification

In order to incorporate the given improvement in Dynam3D, one can alter the architecture by placing a Graph Neural Scene Encoder between the instance representation phase and the 3D Vision-Language Model. The original pipeline uses CLIP to obtain patch-level visual features of RGB-D observations and project them back to 3D space to create patch tokens. FastSAM segmentation of these patches and an instance encoder generate instance tokens of the patches and aggregate them into zone-level tokens depending on spatial proximity.

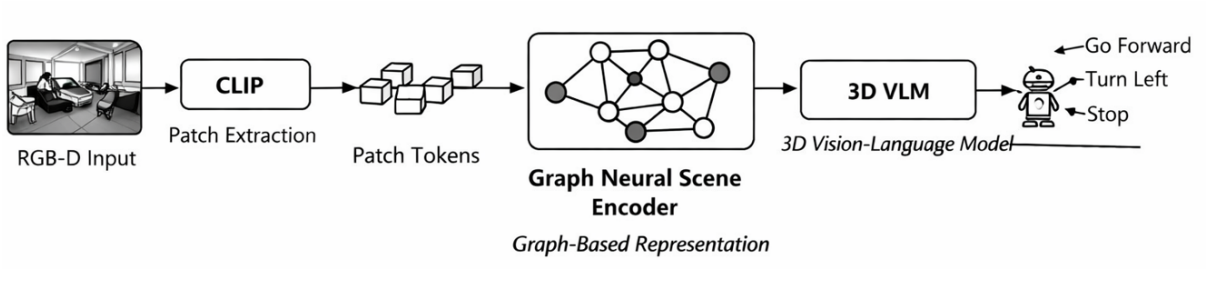

Figure 1: Architecture of the System

The zone aggregating step in the modified architecture is substituted with a graph-based scene representation module. The tokens of each instance are the nodes of a spatial graph, and the relationships between them, including distance, orientation, and co-occurring in the environment, are represented by edges. A Graph Neural Network takes this data in the form of nodes and edges and learns contextual relationships between objects.

3.4 Mathematical / Theoretical Justification

The suggested graph-based scene representation will enhance navigation reasoning by the probability that there are explicit representations of the relationship between objects in the environment. Learning of latent representations in GNNs Node features are gradually updated by exchanging messages between neighbouring nodes, and the model can also learn about spatial dependencies, allowing consideration of aspects like proximity and orientation (Habibpour and Afghah, 2025). This architecture enables the system to model the environment in a semantic manifold, where embeddings of objects that have similar spatial context are similar. Relational propagation is better than simple zone aggregation in terms of representation of features and understanding of context. Also, the gradients are efficiently propagated across

the graph layers in the process of backpropagation, which enables the network to acquire meaningful spatial patterns at a computational cost and scalable scene modelling.

3.5 Expected Benefits

The scene representation offered in the form of the graph is likely to enhance the performance of navigation since it gives an opportunity to perform essential relational reasoning between objects and space areas. The system would be better able to interpret the instructions including the spatial context by explicitly modelling object relationships, which can be expected to bring about increased success rates in navigation. Also, storing environments in the form of graphs may decrease the amount of large storage of features based on zones, which may decrease memory consumption. Generalization can also be improved due to the relational learning ability of Graph Neural Networks, which can enable the navigation model to be more adaptable to previously unknown environments and complicated patterns of instruction.

3.6 Expected Evaluation Plan

The same benchmark datasets used in the original study, as well as R2R-CE and REVERIE-CE, will be used to evaluate the proposed improvement in order to make a fair comparison with the existing methods. These datasets are realistic vision-and-language navigation tasks, which involve spatial reasoning and post-instruction following (Zhang et al. 2025). The measures of performance will be standard measures including Success Rate (SR), Success weighted by Path Length (SPL) and Navigation Error (NE) to measure the accuracy and efficiency of navigation. Other measures, such as memory consumption and the rate of inference, will be examined to measure the computational efficiency. The better model will be compared to the original Dynam3D, and good baselines like NaVid and g3D-LF.

4. Conclusion

Dynam3D is a major step in the field of vision-and-language navigation by providing a hierarchical 3D representation, dynamics of which enhance navigation performance and spatial comprehension. Nevertheless, the high computational complexity, the use of depth sensors, and weak modelling of spatial relationships are limitations that can be improved. This report suggested incorporating a graph-based scene representation system with the help of Graph Neural Networks that would promote relational reasoning among objects. This would enhance a higher level of success in navigation, scalability and generalization, which may result in smoother and more powerful implementation of multimodal navigation systems in practice.

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